F18CF – Linear Algebra

Tutorial Sheet 4: Subspaces

Getting started.

1. Determine whether the following are subspaces of $\mathbb{R}^3$:
   (a) The set S_1 of vectors of the form $(a, 1, 0)^T$ for $a \in \mathbb{R}$;
   (b) The set S_2 of vectors of the form $(a, b, a)^T$ for $a, b \in \mathbb{R}$;
   (c) The set S_3 of vectors $\{(x, y, z)^T \in \mathbb{R}^3 \,:\, 2x - 3y + z = 0\}$;
   (d) The set of vectors $S_4 = \{(x, y, z)^T \,|\, x^2 + y^2 + z^2 \leq 1\}$.

2. Find the dimension and a basis of the solution space of the following homogeneous systems of linear equations:
   (a)
   $$\begin{cases} a + b + 2c &= 0 \\ 2a - b + c &= 0 \\ 3b + 3c &= 0 \end{cases}$$
   (b)
   $$\begin{cases} -a + c + 2d &= 0 \\ -a + b - d &= 0 \\ -b + c + 3d &= 0 \\ a - 2b + c + 4d &= 0 \end{cases}$$

3. Determine the dimension and a basis of the following subspaces:
   (a) the subspace of $\mathbb{R}^3$ spanned by $\vec{u}_1 = (1, 3, 4)^T$, $\vec{u}_2 = (4, 0, 1)^T$ and $\vec{u}_3 = (3, 1, 2)^T$,
   (b) the subspace of $\mathbb{R}^4$ spanned by $\vec{u}_1 = (2, 1, 0, -1)^T$, $\vec{u}_2 = (4, 8, -4, -3)^T$, $\vec{u}_3 = (1, -3, 2, 0)^T$ and $\vec{u}_4 = (1, 10, -6, -2)^T$

4. Determine which of the following sets are subspaces of $\mathcal{P}(\mathbb{R})$, the vector space of all real polynomial functions:
   (a) The subset S_1 of polynomial functions $P \in \mathcal{P}(\mathbb{R})$ such that $P(1) = 0$;
   (b) The subset S_2 of polynomial functions $P \in \mathcal{P}(\mathbb{R})$ such that $P(0) = 1$;
   (c) The subset S_3 of polynomial functions $P \in \mathcal{P}(\mathbb{R})$ such that $P'(0) = 0$.

Problems.

5. **Hyperplanes.** Let $a_1, \ldots, a_n \in \mathbb{R}$ be numbers that are not all zero. We define a *hyperplane* H of $\mathbb{R}^n$ as the set of vectors $(x_1, \ldots, x_n)^T$ of $\mathbb{R}^n$ such that

$$a_1 x_1 + \ldots + a_n x_n = 0.$$

   (a) Show that H is a subspace of $\mathbb{R}^n$.

   (b) What is the dimension of H?

   (c) Find a basis of the hyperplane of $\mathbb{R}^4$ of equation $x_1 + 2x_2 + 3x_3 + 4x_4 = 0$.

6. **New subspaces from old ones.** Let V, W be two subspaces of $\mathbb{R}^n$.

   (a) Show that the intersection $V \cap W$ is again a subspace of $\mathbb{R}^n$.

   (b) Give an example of subspaces such that $V \cap W = \{\vec{0}\}$.

   (c) Is $V \cup W$ always a subspace of $\mathbb{R}^n$?

7. **Sequences satisfying a linear recurrence relation.** We consider the set V of sequences such that for every $n \geq 0$, we have:

$$u_{n+2} = 4u_{n+1} - 4u_n.$$

   (a) Show that V is a subspace of the space $\mathbb{R}^{\mathbb{N}}$ of all real sequences.

   (b) Show that the sequences defined by $v_n = 2^n$ and $w_n = n2^n$ belong to V.

   (c) Are these two vectors linearly independent?

8. **A few examples of subspaces.** We say that a sequence $(u_n) \in \mathbb{R}^{\mathbb{N}}$ is periodic if there exists an integer T such that $u_{n+T} = u_n$ for every $n \in \mathbb{N}$. Similarly, a function $f \in \mathcal{F}(\mathbb{R})$ is periodic if there a real number $T \in \mathbb{R}$ such that $f(x+T) = f(x)$ for every $x \in \mathbb{R}$.

   (a) Show that the subset S_1 of all periodic sequences is a subspace of $\mathbb{R}^{\mathbb{N}}$.

   (b) Show that the subset S_2 of all 2π-periodic functions is a subspace of $\mathcal{F}(\mathbb{R})$.

   (c) Show that the subset S_3 of all periodic functions is *not* a subspace of $\mathcal{F}(\mathbb{R})$.